Advanced Data Science

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Where we are in the course

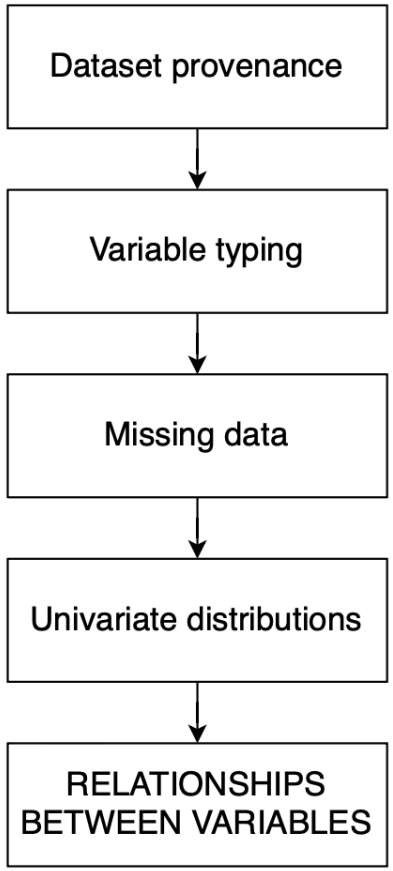
So far we've talked about:

- What does each variable represent?
- How is it measured?
- What does missingness mean?
- What is the distribution?

Now:

- How do variables relate to each other?

numeric–numeric	correlation
categorical–numeric	group comparison
Categorical–categorical	contingency analysis



Three different concepts

Concept	Meaning
Correlation	linear co-movement (Pearson)
Association	general statistical relationship
Dependence	conditional distribution changes

Covariance

$$Cov(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Pearson Correlation

$$\rho = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$$

- Correlation = standardized covariance

$$-1 \leq r \leq 1$$

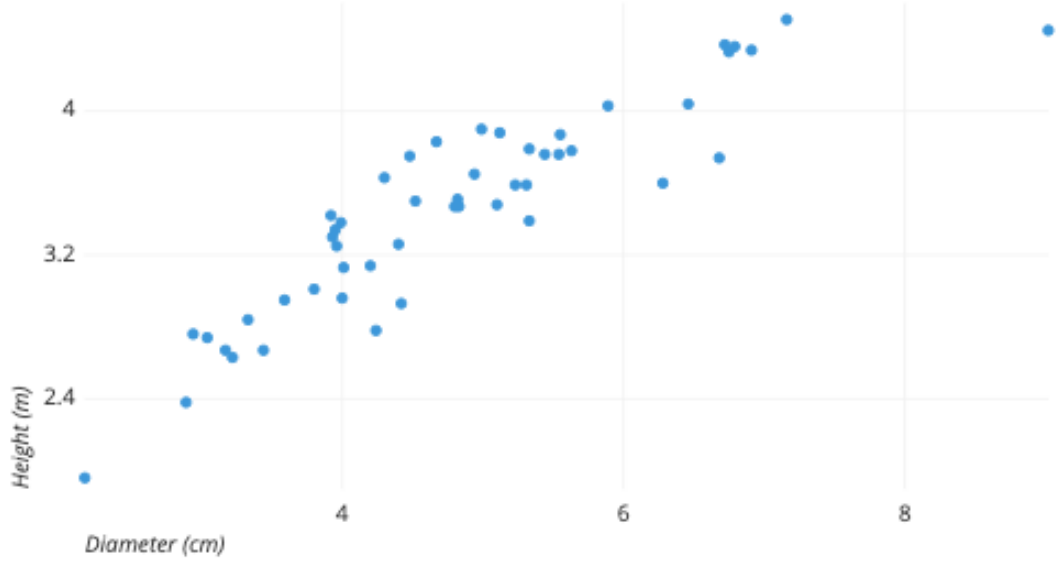
Interpreting correlation

r	Interpretation
0	no linear relationship
0.3	weak
0.6	moderate
0.9	strong

- These thresholds are conventions, not statistical laws.
- This is not a slope.

Always look at the scatterplot

A correlation coefficient without a plot is incomplete.



- numeric variables
- approximately linear relationship
- no strong outliers

https://www.atlassian.com/data/charts/what-is-a-scatter-plot

Spearman Correlation

$$\rho_s = corr(rank(X), rank(Y))$$

Spearman correlation measures **monotonic relationships**

If a relationship is curved but always increasing, which correlation should we use?

Association respecting variable types

numeric	numeric	Pearson / Spearman
categorical	numeric	group comparison
categorical	categorical	contingency analysis

- Relation with variable type.
- Which variables are associated with SalePrice?

Target association analysis

Variable	Type	Measure	Value
Gr Liv Area	numeric	Pearson	0.71
Overall Qual	ordinal	Spearman	0.82
Garage Cars	numeric	Pearson	0.64

- Predictors ↔ SalePrice
- The goal is identify variables that carry information about the target

Multicollinearity

- Multicollinearity occurs when predictors contain overlapping information.
- Predictors correlated with each other
- Predictor ↔ Predictor

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X1 = house size
X2 = number of rooms
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- GarageCars ↔ GarageArea (0.89)
- If two predictors measure almost the same thing, do we need both?

Diagnosing Multicollinearity

How can we detect multicollinearity?

- correlation matrix
- Variance Inflation Factor (VIF)

$$VIF = \frac{1}{1 - R^2}$$

Diagnosing Multicollinearity

VIF = 1
Não há correlação com as outras variáveis. Sem multicolinearidade.
1 < VIF < 5
Há alguma correlação, mas geralmente não é preocupante.
5 ≤ VIF < 10
Multicolinearidade moderada a alta. Convém investigar.
VIF ≥ 10
Multicolinearidade forte. Normalmente considerado problemático.

Practical Takeaways

Three rules for responsible correlation analysis:

- Always visualize the relationship
- Choose the association measure that matches the variable type
- Check correlations among predictors

Correlation is easy to compute, but responsible interpretation requires context, visualization, and careful thinking.

Obrigado!